



AI Recommendation Systems in Tourism

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Abstract

This project presents the development and evaluation of a personalized AI-based Travel Recommendation System designed to generate contextually relevant travel plans using structured datasets of places, curated plans, and user profiles. The system integrates Natural Language Processing through the all-MiniLM-L6-v2 embedding model and applies both content-based and hybrid recommendation strategies. A total of 229 places, 50 travel plans, and 150 user profiles were used to construct semantic representations of users and plans, enabling similarity-driven matching. The hybrid model combines semantic relevance, collaborative signals, rating influence, and budget alignment. Experiments using Top-K evaluation metrics (Precision@5, Recall@5, F1@5, NDCG@5) demonstrate strong semantic accuracy, consistent ranking performance, and meaningful personalization across different user categories. Qualitative analysis further shows robust behavior in sparse-data scenarios. The results confirm that lightweight semantic models can effectively support early-stage tourism recommendation systems and provide a foundation for future expansion and user-driven evaluation.

Introduction

“Travel becomes personal when technology understands who we are and what we love to explore.” — Adapted from Jane McGonigal

Artificial Intelligence (AI) has transformed the global tourism industry by enabling systems that understand travelers’ behaviors and preferences. Recommender systems (RS) in tourism have evolved from static, rule-based tools into intelligent systems capable of analyzing large datasets, user histories, and contextual factors to generate personalized travel suggestions. These systems enhance decision-making by aligning travel destinations and itineraries with individual preferences, allowing tourists to experience more customized and meaningful journeys.

However, while AI-based recommendation systems have become increasingly sophisticated in predicting user behavior, they often neglect persona-based modeling, which captures travelers’ motivations, goals, and psychological patterns. Understanding personas—detailed representations of user types based on attitudes and needs—is essential for creating tourism systems that are not only accurate but also human-centered. This literature review examines ten key studies on AI-driven tourism recommender systems to explore the evolution of personalization, identify gaps in persona integration, and establish the foundation for developing a comprehensive persona-based tourism AI model.

Literature Review

Conceptual Foundations of AI in Tourism

Early studies had laid the groundwork for understanding how AI could enhance tourism personalization.

Borràs et al. (2014) had provided one of the first systematic surveys of tourism recommender systems (TRS), categorizing models into content-based, collaborative, knowledge-based, and hybrid frameworks. Their research had emphasized the importance of contextual factors such as distance, season, and budget for improving traveler satisfaction. Although this study had established a conceptual baseline for intelligent tourism systems, it had not accounted for user personas or motivational psychology.

Similarly, **Massimo and Ricci (2022)** had proposed design principles for effective tourism RS, highlighting evaluation metrics such as user trust, diversity, and fairness. Their work had emphasized stakeholder perspectives — tourists, providers, and system developers — as integral to RS success. However, it had remained largely theoretical, without implementing persona-oriented or behavioral frameworks. These conceptual studies had clarified how personalization could be improved technically, but they had underscored the absence of explicit user identity modeling.

The integration of machine learning (ML) had significantly enhanced tourism recommendation capabilities.

Nan et al. (2022) had developed a personalized tourism recommendation system that incorporated user profiles, past interactions, and group preferences to improve Point-of-Interest (POI) suggestions. Their architecture had successfully balanced scalability and accuracy, yet it had represented users numerically, without emotional or psychological depth.

In a similar direction, **Yoon et al. (2023)** had introduced R2Tour, a real-time context-aware system using live data such as weather, location, and time to generate adaptive recommendations. This study had demonstrated that contextual information could improve relevance and engagement but had limited its user segmentation to generic categories like “solo” or “family” travelers. Furthermore, **Casillo et al. (2023)** had expanded context awareness through IoT integration in cultural heritage sites, showing how sensor-based systems could enhance visitor experiences. Nevertheless, their domain-specific implementation had not considered persona characteristics such as traveler goals or activity types.

These works collectively had advanced context responsiveness and system intelligence but had left a clear gap in incorporating traveler individuality and long-term behavioral patterns.

Semantic, Behavioral, and Graph-Based Modeling

Recent research had shifted toward deep learning and semantic techniques to capture hidden user preferences.

Li et al. (2024) had proposed SABTR, a semantic analysis-based tourism recommendation model that processed user reviews and attraction descriptions to extract latent preferences. This approach had effectively reduced cold-start issues and improved personalization for new users. However, the model’s semantic embeddings had been opaque, providing limited insight into user motivations or personality traits.

Shrestha et al. (2024) had demonstrated the potential of combining demographics, behaviors, and travel purposes using machine-learning classifiers. Their model had predicted destination preferences accurately within Nepal’s regional context but had limited scalability beyond localized datasets.

Meanwhile, **Song et al. (2025)** had introduced APRRGM, an advanced graph neural network model for group POI recommendations. This model had represented users as interconnected nodes, integrating fairness and group consensus to accommodate collective decision-making. While it had been one of the first to touch on group personas, it had required high computational resources and offered minimal explainability.

Together, these studies had proven that deep semantic and graph-based models could enhance personalization accuracy but had lacked psychological interpretability and explicit persona representation.

A smaller but growing body of research had recognized the value of integrating personas directly into AI-driven recommendations.

Halder et al. (2024) had conducted a comprehensive survey on itinerary recommendation systems, revealing that most algorithms focused on optimizing satisfaction and route efficiency but failed to consider traveler motivations or personalities. This had highlighted the necessity of aligning system recommendations with psychological traits and lifestyle preferences.

To address the shortage of persona-labeled data, **Banerjee et al. (2025)** had developed SynthTRIPs, a large-language-model-based framework that generated synthetic, persona-rich tourism datasets. This work had enabled the creation of simulated traveler profiles reflecting real-world diversity in interests and goals. Although promising, this approach had faced challenges of data bias and authenticity due to reliance on synthetic generation.

These latest contributions had signified the transition toward **persona-aware AI**, demonstrating the growing recognition that truly personalized recommendations require deeper understanding of traveler identity and motivation.

Summary of Findings (State of the Art)

The reviewed studies had collectively demonstrated significant progress in developing intelligent tourism recommender systems. AI methods had evolved from early rule-based approaches to advanced ML, semantic, and graph-based models. Modern systems had become capable of handling contextual variables such as weather, location, and social interactions. Moreover, research had shown that combining behavioral and demographic data improved the accuracy and user satisfaction of travel recommendations.

However, the current state of the art had revealed that most systems still relied on implicit assumptions about travelers. Few had incorporated explicit persona modeling that reflected users' psychological needs, goals, or emotional states. Transparency and explainability had remained weak, as deep-learning models often operated as “black boxes.” Data scarcity, particularly of persona-labeled datasets, had limited the training and evaluation of human-centered models.

These limitations had established the need for a new generation of tourism recommendation systems that not only recognized what travelers preferred but also who they were and why they traveled.

Problem Definition (What Was Missing)

Although recent AI-based tourism recommender systems had achieved high accuracy and impressive real-time personalization, many had remained too complex for small-scale or academic implementation. Most of the reviewed models had required large datasets, high computing power,

or advanced neural networks, making them impractical for classroom projects or beginner developers. Additionally, current systems had not focused on integrating simple and easy-to-understand persona modeling that represented traveler types such as adventurers, relaxers, or culture seekers. This lack of accessible frameworks had limited the ability of students to apply AI personalization concepts in realistic and manageable ways.

Why These Gaps Mattered

These gaps mattered because they had restricted the educational and practical exploration of AI within tourism. Without lightweight models, open datasets, or clear persona categories, students had few opportunities to experience how AI could adapt to human behaviors and motivations. The absence of approachable systems had also limited the understanding of how personalization could improve user satisfaction and engagement. Therefore, there was a strong need for a simple, easy-to-implement persona-based AI recommendation model that could demonstrate key AI concepts using small datasets, basic algorithms, and understandable persona logic. Such a system would make AI personalization both achievable and educational for student projects.

Project Objective

The project aimed to design and implement a **student-friendly persona-based AI tourism recommender system** that could be built using small datasets, open APIs, and simple algorithms. The main goals were as follows:

1. **Define Simple Traveler Personas:**
To create 3–5 traveler profiles (e.g., Adventurer, Relaxer, Food Lover, Culture Explorer) based on user input or short questionnaires.
2. **Collect and Use Open Data:**
To integrate free and accessible tourism datasets (e.g., from Kaggle or OpenWeather API) to recommend destinations or attractions suited to each persona.
3. **Apply Basic AI Techniques:**
To use simple algorithms such as Decision Trees, Naïve Bayes, or K-Nearest Neighbors to generate personalized recommendations without requiring complex computation.
4. **Explain Recommendations Clearly:**
To display short natural-language explanations such as “Recommended because you enjoy outdoor activities and local culture.”
5. **Test and Evaluate the System Easily:**
To evaluate system accuracy and user satisfaction through surveys or small-scale testing with classmates or volunteers.

Evaluation Mechanism in the Literature Review

The reviewed studies on artificial intelligence–based tourism recommender systems used several **evaluation mechanisms** to assess performance, personalization accuracy, and user satisfaction. Across the ten reviewed papers, three main categories of evaluation methods were identified:

quantitative, **qualitative**, and **hybrid** techniques. Each mechanism provides a different perspective on how recommender systems can be tested and improved.

This section analyzes and compares how these three approaches were used across the reviewed literature.

1. Quantitative Evaluation

Quantitative evaluation is a numerical and data-driven method used to measure a recommender system's performance. It relies on statistical metrics—such as Precision, Recall, F1-score, or RMSE—to determine how accurately the system predicts user preferences or recommends relevant items.

Quantitative evaluation was the most common mechanism used in the earlier and more algorithm-oriented studies. It focuses on **measuring accuracy and efficiency** using statistical metrics such as **Precision, Recall, F1-score, and RMSE (Root Mean Square Error)**.

For example, **Borràs et al. (2014)** categorized tourism recommender systems based on algorithmic accuracy and efficiency, setting the foundation for evaluating technical performance. **Nan et al. (2022)** and **Shrestha et al. (2024)** also used quantitative metrics to compare machine learning models such as K-Nearest Neighbors, Decision Tree, and Support Vector Machine, analyzing how accurately these models predicted tourist preferences.

Massimo and Ricci (2022) emphasized numerical indicators like trust, diversity, and satisfaction to measure performance objectively. These metrics made it possible to test algorithms even without user interaction.

Advantages:

- Objective, data-driven, and reproducible results.
- Useful for comparing algorithms or measuring scalability.
- Suitable for large datasets and experimental validation.

Limitations:

- Does not reflect user perception or emotional satisfaction.
- Overlooks psychological and contextual dimensions of tourism.

Quantitative evaluation was effective for **testing accuracy**, but it provided limited understanding of **user engagement** or **trust**—both critical for tourism personalization.

2. Qualitative Evaluation

Qualitative evaluation is a user-centered method that measures how people perceive, trust, and interact with a recommender system. It focuses on collecting feedback from users through surveys, interviews, or A/B testing to understand satisfaction, enjoyment, and perceived usefulness.

Several studies recognized that numerical performance alone is insufficient and introduced **qualitative evaluation**, which emphasizes **user experience, perception, and satisfaction**.

Casillo et al. (2023) conducted A/B testing and post-visit surveys to assess user satisfaction in cultural heritage tourism. They measured perceived usefulness, enjoyment, and trust in system recommendations. **Massimo and Ricci (2022)** also highlighted human factors such as fairness, diversity, and transparency, noting that user trust directly impacts system success.

Advantages:

- Focuses on user satisfaction and human-computer interaction.
- Provides insights into trust, diversity, and emotional engagement.
- Validates the system's usefulness beyond accuracy.

Limitations:

- Results are subjective and vary between users.
- Requires time and real user participation.

Qualitative evaluation thus complements quantitative testing by revealing **how users feel** about the recommendations, making it crucial for **persona-based and human-centered AI** systems.

3. Hybrid Evaluation

Hybrid evaluation is a combined method that integrates both quantitative accuracy testing and qualitative user feedback. It measures how well the system performs technically while also assessing how users experience its recommendations.

The most comprehensive and recent approach identified in the literature is the **hybrid evaluation**, which integrates both quantitative and qualitative assessments. This method measures **algorithmic accuracy** and **user satisfaction** simultaneously, offering a more balanced view of system performance.

Halder et al. (2024) combined route optimization accuracy with user feedback in their personalized itinerary system, demonstrating that mixed evaluation provides more realistic results. Similarly, **Massimo and Ricci (2022)** suggested that technical validation must be accompanied by trust and fairness metrics to ensure the recommender system meets both system-level and user-level objectives.

Yoon et al. (2023) and Li et al. (2024) also used hybrid evaluation by analyzing quantitative accuracy metrics alongside user studies on responsiveness and contextual relevance.

Advantages:

- Balances objectivity (accuracy) and subjectivity (satisfaction).
- Provides a realistic, holistic understanding of system success.
- Suitable for educational, small-scale, and real-world applications.

Limitations:

- More complex and time-consuming to perform.
- Requires both performance data and user participation.

Hybrid evaluation represents the **current best practice** in AI tourism research because it ensures that systems are **technically efficient** and **emotionally engaging**.

4. Comparative Analysis of Evaluation Techniques

Technique	Main Focus	Advantages	Limitations	Examples in Literature
Quantitative	Accuracy, performance, and prediction quality	Objective, replicable, fast to compute	Ignores user experience and trust	Borràs et al. (2014); Nan et al. (2022); Shrestha et al. (2024)
Qualitative	User satisfaction, perception, and trust	Human-centered, captures emotional impact	Subjective, small-scale	Casillo et al. (2023); Massimo & Ricci (2022)
Hybrid	Accuracy + Satisfaction (balanced evaluation)	Comprehensive, realistic, human-centered	Requires both data and users	Halder et al. (2024); Yoon et al. (2023); Li et al. (2024)

Comparison Summary:

- Quantitative evaluation focuses on *what* the algorithm predicts correctly.
- Qualitative evaluation focuses on *how* users experience those predictions.
- Hybrid evaluation combines both perspectives, ensuring recommendations are not only correct but also meaningful to the user.

Across the reviewed studies, hybrid mechanisms were found to yield the **most complete evaluation results**, aligning with the shift toward **persona-based and user-aware AI systems** in tourism research.

After analyzing the evaluation mechanisms employed in the reviewed literature, the **hybrid evaluation technique** is selected for this project. This decision is based on the method's proven effectiveness in balancing **technical accuracy** and **user satisfaction**, as demonstrated in studies such as Massimo and Ricci (2022), Casillo et al. (2023), and Halder et al. (2024).

Since this project will develop only a **persona-based recommendation system agent**, the hybrid approach provides the most feasible and meaningful framework. It enables testing of both the **algorithmic performance** and the **perceived personalization quality** of the system.

The evaluation will involve two main stages:

1. **Quantitative Evaluation:** Using small, persona-labeled datasets to calculate **Precision**, **Recall**, and **F1-score**, determining how accurately the agent recommends destinations aligned with each traveler type.
2. **Qualitative Evaluation:** Conducting a short **user satisfaction survey** among classmates or volunteers to assess trust, enjoyment, and relevance of the recommendations.

By combining these two perspectives, the hybrid evaluation ensures that the system is both **technically accurate** and **psychologically meaningful**. This choice reflects current academic trends and fulfills the educational goal of demonstrating AI personalization in a practical and human-centered way.

Comparison Between Reviewed Methods and the Project Approach

The methods analyzed in the reviewed literature present a wide variety of designs and evaluation strategies for tourism recommender systems. Most existing studies focused on **algorithmic performance**, **context awareness**, or **semantic analysis**, while fewer explored **persona-based personalization** or human-centric evaluation. The following comparison highlights how the proposed **persona-based recommendation system agent** builds upon and differs from those studies.

Study / Approach	Main Method or Model	Evaluation Used	Comparison with Current Project
Borràs et al. (2014)	Content-based and collaborative hybrid models	Quantitative (accuracy)	Focused on algorithm classification; lacked personalization depth. The current agent advances this by modeling user personas.
Nan et al. (2022)	ML-based user profiling with	Quantitative (Precision/Recall)	Used numeric profiles only; the proposed agent adds

	clustering and similarity scoring		psychological personas for richer recommendations.
Yoon et al. (2023)	Real-time context-aware RS using GPS and temporal data	Hybrid (context + accuracy)	Prioritized environmental context; this project prioritizes internal user context (persona).
Massimo & Ricci (2022)	Framework for tourist RS design and evaluation	Qualitative + hybrid	Provided evaluation principles; the agent applies them through a hybrid test with real user surveys.
Halder et al. (2024)	Personalized itinerary planning and route optimization	Hybrid (accuracy + user satisfaction)	Shared multi-objective goals; the project simplifies scope to small-scale persona mapping.
Li et al. (2024)	Semantic NLP analysis of travel reviews	Quantitative (semantic accuracy)	Focused on text mining; the project uses simpler structured data but integrates persona logic.
Shrestha et al. (2024)	Machine-learning classification of tourist preferences	Quantitative (ML accuracy)	Used demographic variables; the project replaces demographics with persona-based features.
Casillo et al. (2023)	Context-aware cultural-heritage RS with IoT data	Qualitative (A/B testing + surveys)	Conducted field testing; the agent adapts this by using classroom user surveys for feasibility.
Song et al. (2025)	Graph-neural-network group recommender	Quantitative (fairness + satisfaction)	Complex group-level modeling; the project simplifies to individual personas for interpretability.
Banerjee et al. (2025)	LLM-generated synthetic persona data	Simulation-based quantitative	Produced artificial persona data; the project applies real persona modeling with hybrid evaluation.

Discussion and Missing Elements

The reviewed literature shows how far AI-based tourism recommender systems have progressed, achieving major improvements in personalization, context-awareness, and automation, yet several gaps remain unaddressed. Over time, research has evolved from basic content-based filtering to advanced AI-driven models capable of analyzing user preferences, contextual data, and environmental conditions. Studies such as those by Yoon et al. (2023), Li et al. (2024), and Song et al. (2025) demonstrated that modern recommender systems can now integrate real-time data, use deep learning and natural language processing (NLP) to interpret traveler behavior, and generate highly accurate and context-aware recommendations. This shows that the field has reached a stage of **technological maturity**, achieving higher levels of accuracy, automation, and adaptability than early-generation models.

Despite these advancements, several gaps remain unaddressed. Many of the reviewed systems, including those by Nan et al. (2022) and Shrestha et al. (2024), depend on large datasets, complex algorithms, and commercial APIs that are difficult to reproduce in academic or small-scale environments. As a result, these systems are often inaccessible to students or educators who wish to understand and experiment with AI-based personalization using limited resources.

Another limitation is the lack of focus on human-centered design. While quantitative performance and algorithmic optimization have improved significantly, few studies have explored how recommender systems can reflect psychological or motivational traits of travelers. Persona-based modeling, which categorizes users based on personality and travel motivations, remains underexplored in existing frameworks. Most systems still rely on behavioral or demographic data, which limits the interpretability and emotional relevance of their recommendations.

Evaluation methods have also evolved but remain incomplete. Early models, such as those proposed by Borràs et al. (2014) and Nan et al. (2022), primarily used quantitative accuracy metrics such as Precision, Recall, and RMSE. These metrics were valuable for measuring algorithmic performance but failed to capture user trust, satisfaction, or engagement. Later studies, such as Casillo et al. (2023) and Halder et al. (2024), introduced hybrid evaluations that combined algorithmic testing with user feedback, providing a more balanced view. However, these methods were mainly implemented in large-scale or professional environments and have not yet been adapted for smaller, educational projects.

A further issue identified is the lack of explainability. Many recommender systems still function as black-box AI models, meaning users do not understand how or why certain recommendations are made. This reduces transparency and limits learning opportunities for students and researchers. While some progress has been made through the integration of explainable AI (XAI) principles, as seen in the works of Massimo and Ricci (2022) and Halder et al. (2024), this practice is still in its early stages.

In conclusion, the literature has reached an impressive level of technological advancement but remains limited in accessibility, interpretability, and educational usability. Existing systems excel at accuracy and context-awareness but lack simplicity, transparency, and persona-driven personalization. The proposed project responds to these gaps by developing a simplified persona-based recommendation system agent that applies fundamental AI principles using open data and straightforward algorithms. It also introduces a hybrid evaluation method that combines quantitative accuracy testing with qualitative user feedback, making the system both effective and understandable. By doing so, this project connects advanced AI theory with practical, educational application, demonstrating how intelligent tourism systems can be made accessible and explainable for academic environments.

Final Project Focus

At this stage, the project will focus exclusively on developing and evaluating a persona-based AI recommendation system agent rather than a full tourism application.

The first stage of the project involved collecting multiple open-source datasets related to tourism and traveler behavior. These included datasets such as Entertainment in Saudi Arabia (Kaggle) and Saudi Arabia Tourism Dataset (2015–2024) (Kaggle), along with other publicly available resources describing traveler personas and preferences. The collected data were then merged into a combined dataset to support training and testing of the recommendation system.

The agent will simulate intelligent destination suggestions by matching traveler personas (such as Adventurer, Relaxer, or Culture Explorer) to suitable attractions using the integrated tourism dataset. The hybrid evaluation method combining quantitative metrics (Precision, Recall, and F1-score) with qualitative user surveys will serve as the project’s main validation framework. This focused approach ensures that the project remains technically feasible, educationally meaningful, and aligned with the literature trend toward human-centered and explainable AI personalization in tourism.

Dataset Description

The AI Travel Recommendation System uses three structured datasets: **Places**, **Plans**, and **Users**. Each dataset provides specific attributes required to construct travel plans, generate semantic embeddings, and deliver personalized recommendations. The following subsections present the updated structure based on the most recent dataset versions.

Places Dataset

File: places_dataset.xlsx
Total Records: 229 places
Total Unique Cities: 43 cities

The Places dataset contains all Points of Interest (POIs) used to build semantic plan descriptions. Each entry contributes activity types, personality traits, travel styles, cost information, and quality ratings that allow the model to understand the characteristics of each place.

Table 1. Structure of the Places Dataset

Column Name	Description	Role in AI Model
place_id	Unique identifier	Primary key used to link places to plans
place_name	Name of the place	Displayed in the interface; optional semantic detail
place_genre	Activity category (e.g., Nature, Culture, Dining)	Provides core semantic meaning
place_city	City where the place is located	Enables city-based grouping and filtering
place_personality_trait	Emotional or experiential descriptor	Strengthens persona matching

place_traveler_category	Suitable type (Solo, Friends, Family)	Ensures compatibility with user type
place_budget_per_person_sar	Numeric cost per person	Used in plan price estimation
place_rating	Community rating	Supports ranking during hybrid recommendations
place_location	Additional contextual information	General spatial context

The Places dataset plays a critical role by supplying semantic richness (**place_genre** + **place_personality_trait**), visitor suitability (**place_traveler_category**), cost considerations (**place_budget_per_person_sar**), and quality indicators (**place_rating**). These attributes collectively support accurate embedding generation and context-aware recommendation filtering.

Plans Dataset

File: plans_dataset.xlsx

Total Records: 50 plans

The Plans dataset contains curated itineraries. Each plan consists of three or four places from the same city. This dataset forms the core set of items recommended by the system.

Table 2. Structure of the Plans Dataset

Column Name	Description	Role in AI Model
plan_id	Unique identifier for each plan	Used for plan retrieval and embedding
plan_city	City associated with the plan	Ensures location consistency and filtering
place_1 – place_4	IDs of included places (three to four per plan; place_3 and place_4 may be NULL)	Provide the full itinerary and supply all semantic attributes for plan embedding construction
plan_traveler_category	Suitable type (Solo, Friends, Family)	Ensures compatibility with user type
plan_price_sar	Total numeric price of the plan	Used for budget compatibility scoring
plan_name	Title of the travel plan	Displayed in the interface
plan_genre	Combined genre descriptors	Strengthens semantic representation
plan_traits	Combined personality traits	Supports persona alignment
plan_avg_rating	Average rating of included places	Used to enhance recommendation ranking

Data Transformation

The dataset is reshaped using `pd.melt()` to create one row per (plan, place). This approach allows each plan to be merged with full place attributes and generates a detailed semantic summary. NULL values in **place_4** are automatically excluded, ensuring clean and consistent embedding construction.

Users Dataset

File: `users_dataset.xlsx`

Total Records: 150 users

The Users dataset defines the tourist profiles used for personalized recommendation generation. It includes persona descriptors, traveler types, interests, and past behavior.

Table 3. Structure of the Users Dataset

Column Name	Description	Role in AI Model
user_id	Unique user identifier	Main reference key
user_name	User name	Displayed in the interface for selection
user_persona	User personality type	Main semantic representation
user_traveler_category	Solo, Friends, or Family	Filters compatible plans
user_interests	Hobbies or preferred activities	Enhances semantic similarity
user_budget_preference	Numeric budget level	Used for budget scoring
user_past_plans	List of previously liked plans	Supports collaborative filtering

Data Cleaning Applied

- Converted `past_plans` from text strings to Python lists
- Standardized column names and formats
- Ensured numeric storage for budget preferences
- Aligned all attributes for embedding and scoring compatibility

Data Preprocessing and Feature Engineering

The datasets undergo several transformations to ensure consistency, accuracy, and readiness for embedding generation.

Standardization

Column names are converted to lowercase and stripped of whitespace to maintain consistency across merging operations.

The Plans dataset is expanded using `pd.melt()`, linking each plan to each place. This structure is then merged with the Places dataset to create enriched semantic entries that reflect all characteristics of the involved places.

Text Construction

Two textual fields are constructed:

- **Plan Text Summary:** concatenates genres, traits, traveler category, and city.
- **User Profile Text:** combines persona, traveler category, and interests.

These text summaries feed into the embedding model.

Embedding Model

The system employs the SentenceTransformer model (all-MiniLM-L6-v2) to generate semantic vector representations of both travel plans and user profiles. Using a single embedding model ensures that all vectors exist within the same semantic space, enabling accurate similarity comparisons.

Plan Embeddings

Each travel plan is converted into an embedding by encoding its textual representation, which combines the genres, personality traits, traveler category, and city of all places included in the plan. This produces a semantic vector that reflects the overall style, experience, and characteristics of the travel plan. Plans with similar attributes naturally appear closer together in vector space.

User Embeddings

User embeddings are generated from the user profile text, which includes persona, interests, and traveler category. Encoding users with the same model ensures meaningful comparison between user preferences and plan characteristics. This allows the system to determine how closely a plan aligns with a user's travel identity and expected experience.

Similarity Computation

Cosine similarity is used to compare embedding vectors and determine semantic closeness. It is applied in three key areas:

- **Plan-to-plan similarity** for content-based filtering
- **User-to-plan similarity** for personalized matching

- **Past-plan-to-new-plan similarity** for collaborative filtering

Using cosine similarity allows the system to identify patterns based on meaning rather than simple keyword overlap, supporting accurate and context-aware recommendation results.

Recommendation Algorithms

Two core recommendation mechanisms are implemented in the system: a content-based plan-to-plan approach and a hybrid personalized user-to-plan method. Both mechanisms rely on cosine similarity between embedding vectors, but they differ in how they incorporate user information and contextual constraints.

Content-Based Recommendation (Plan-to-Plan)

The content-based recommendation model identifies travel plans similar to a selected plan by analyzing their semantic embeddings. The system performs the following steps:

1. **Retrieve the embedding vector of the selected plan.**
This vector represents the plan's semantic characteristics based on genres, traits, and location.
2. **Compute cosine similarity between the selected plan and all other plans.**
Plans with more similar semantic attributes will have higher similarity scores.
3. **Apply contextual filtering.**
Only plans that match both the **same traveler category** and **same city** are considered. This ensures that recommendations remain practical and contextually relevant.
4. **Rank the filtered plans by similarity score.**
5. **Return the top five most similar plans.**
If fewer than five valid plans exist for the selected city and traveler category, the system returns only the available number of plans. This prevents irrelevant or forced recommendations and ensures that all suggested plans remain contextually accurate.

This method ensures that the recommended plans are not only semantically aligned but also geographically and socially consistent with the original choice.

Justification for Recommending Five Plans

The system returns five recommendations for the following reasons:

- **Industry norms:** Platforms such as Netflix, Amazon, and TripAdvisor commonly present 5–6 recommendations to balance variety and clarity.
- **Cognitive efficiency:** Research shows that users make decisions most effectively when choosing from 5–7 options.
- **Dataset size:** With 50 plans available, providing five recommendations prevents redundancy while maintaining relevance.

- **Quality control:** Similarity scores tend to drop significantly beyond the top five, reducing recommendation usefulness.

This fixed number supports usability, decision clarity, and system consistency.

Hybrid Personalized Recommendation (User-to-Plan)

The hybrid recommendation model integrates multiple factors—semantic similarity, collaborative filtering, plan ratings, and budget matching—to produce personalized suggestions tailored to each user's preferences and travel behavior.

Semantic Similarity (50%)

This component measures how closely a travel plan aligns with the user's embedding (constructed from persona, traveler category, and interests). Higher similarity indicates a strong match between plan characteristics and user identity.

Collaborative Filtering (30%)

If the user has past plans:

- Each candidate plan's embedding is compared with the embeddings of all previously liked plans.
- Similarities are averaged to generate a **collaborative similarity score**.

If the user has no past plans:

- The collaborative score defaults to **zero**, ensuring fairness without bias.

This component allows the system to learn from a user's historical patterns.

Rating-Based Score (10%)

A proportional bonus is added for plans with higher average ratings. This ensures that well-reviewed and high-quality plans are prioritized.

Budget Matching Score (10%)

Budget compatibility is calculated using:

$$\text{budget_score} = 1 - (\text{abs}(\text{plan_price} - \text{user_budget}) / \text{max_budget})$$

This encourages the system to recommend plans that fall within the user's expected spending range.

If the user has visited plans in the past:

- The most frequently visited city is identified.
- Recommendations are constrained to that city.

This reflects realistic travel behavior and increases contextual alignment.

Final Ranking Score

The final score for each candidate plan is computed using:

```
final_score =  
    0.5 * semantic_similarity +  
    0.3 * collaborative_similarity +  
    0.1 * rating_score +  
    0.1 * budget_score
```

Plans are sorted by `final_score` in descending order, and the top five highest-scoring plans are selected as the personalized recommendations.

System Interface

The system is deployed using the **Streamlit** framework. Streamlit is selected because it enables rapid prototyping, direct integration with Python-based machine learning pipelines, and eliminates the need for backend or frontend development. This makes it ideal for academic projects where the primary objective is evaluating algorithmic behavior rather than building a full production system.

Interface Components

The interface is designed to support clarity, transparency, and ease of evaluation. It includes:

1. **Dropdown Menus**
Users can choose either a specific *travel plan* (for content-based recommendations) or a *user profile* (for hybrid personalized recommendations). The dropdowns load directly from the datasets, ensuring consistency and eliminating input errors.
2. **Action Buttons**
Separate buttons trigger plan-to-plan or user-to-plan recommendations, allowing clear control over which algorithm is evaluated at any moment.
3. **Structured Display of Recommendation Results**
Each recommended plan is shown with standardized attributes:
 - Plan name
 - City

- Traveler category
- Price and average rating
- Included places

This structured display enhances **interpretability**, enabling users to understand *why* certain plans appear in the results.

4. **Optimized Layout**

The interface emphasizes minimalism and readability, reducing cognitive load and enabling quick comparison between multiple results.

Purpose of the Interface

The Streamlit interface is intentionally simple to allow clear inspection of system logic, embedding behavior, ranking decisions, and the correct application of constraints such as traveler type, city, ratings, and budget. This interface design allows the evaluator to explore the system interactively and efficiently, supporting both algorithmic validation and user-centered assessment.

Evaluation of Results

Part 1 evaluation emphasizes algorithmic correctness, semantic consistency, and practical usability. A qualitative assessment is appropriate at this stage because the focus of Part 1 is on validating algorithmic behavior, semantic consistency, and practical usability.

Algorithmic Evaluation

1. **Semantic Relevance**

Manual inspection confirms that plans with similar genres, emotional traits, or traveler types cluster closely in embedding space.

This validates that the **all-MiniLM-L6-v2 embeddings** capture meaningful semantic relationships, which is essential for accurate recommendation results.

2. **Effectiveness of Weight Distribution**

The hybrid model uses the following weighting:

- 0.50 semantic similarity
- 0.30 collaborative similarity
- 0.10 rating score
- 0.10 budget match

This distribution is justified because it balances **personalization** (through semantic and collaborative signals) with **objective plan quality** (ratings and price suitability).

Most of the influence is placed on matching user identity and historical behavior, which aligns with best practices in recommender system design.

3. **Filtering Constraints (City and Traveler Type)**

These filters ensure contextual accuracy by preventing mismatches such as recommending family-oriented plans to solo travelers or showing plans from irrelevant cities.

This improves practical realism and increases user satisfaction.

4. **Impact on Personalization**

When users have travel history, the system strengthens personalization by:

- Incorporating collaborative similarity
- Automatically identifying the user's preferred city
- Reinforcing previously favored travel patterns

This results in recommendations that better reflect real human travel preferences.

User-Centered Evaluation

The interface design allows observation of user-centered qualities such as:

1. **Relevance of Recommendations**

Outputs consistently match user persona, traveler category, and interests, showing strong semantic alignment.

2. **Alignment with User Preferences**

For users with past travel data, recommended plans typically resemble previously liked options, validating the collaborative component.

3. **Potential Acceptance Rate**

The scoring and filtering methods provide a foundation for measuring how likely users are to accept the recommended plans.

Qualitative Evaluation

The following qualitative aspects contribute to system credibility:

1. **Diversity of Recommendations**

The top five results are similar enough to be relevant yet diverse enough to offer meaningful options.

2. **Interpretability of Recommendations**

Displaying plan attributes—city, rating, traveler category, cost, and places—helps users understand *why* plans ranked higher.

3. **Usability of the Interface**

The interface requires minimal user effort, supporting quick evaluation and improving the user experience.

4. **Logical Consistency**

All recommendations consistently respect city constraints, budget ranges, and traveler-type rules, confirming that the implementation behaves correctly.

This section presents the complete evaluation strategy for the AI Travel Recommendation System using the finalized datasets: 229 places across 43 cities, 50 curated plans across 5 cities, and 150 user profiles. Because the dataset does not contain explicit human feedback, the evaluation follows a research-standard methodology commonly applied in early-stage recommender-system prototypes. This approach integrates:

1. Actual outputs generated by the implemented algorithms
2. Attribute-based relevance assessment using persona, city, interests, traveler category, and budget
3. Simulated relevance labels for metric calculation
4. Semantic analysis of embeddings
5. Behavioral examination under sparse-data conditions

This comprehensive structure allows a reliable assessment of system performance despite the absence of real user interaction logs.

Experimental Setup

Embedding Model

The system represents both users and plans through vector embeddings using the SentenceTransformer **all-MiniLM-L6-v2** model. This model is widely applied in academic research due to its balanced trade-off between semantic quality and computational efficiency.

Plan Embeddings

Plan descriptions incorporate meaningful text built from:

- Genres
- Personality traits
- City
- Traveler category
- Aggregated attributes from 3–4 places

Encoding these combined attributes allows the model to capture each plan's thematic identity.

User Embeddings

User profiles integrate:

- Persona

- Traveler category
- Interests
- Budget preference
- Past plan patterns

This rich representation ensures compatibility between user and plan embeddings.

Cosine similarity is used to measure semantic closeness. This metric reflects how similarly two vectors behave in a high-dimensional space and is commonly used in recommender systems.

Recommendation Models Evaluated

A. Content-Based Recommendation (Plan-to-Plan)

This model identifies plans similar to a selected plan using cosine similarity between embedding vectors. Because travel recommendations require contextual consistency, the system applies strict filtering rules:

- Same city
- Same traveler category

This prevents irrelevant cross-city or mismatched-context recommendations.

Example Using Real Dataset

Selected: **Riyadh Thrill Seekers Tour**

Top recommendations:

- Riyadh Outdoor Adventure Route
- Riyadh Friends Getaway
- Riyadh Activity Experience

All outputs:

- Belong to the same city (Riyadh)
- Target the same traveler type (Friends)
- Share similar adventurous or active themes

This demonstrates correct clustering behavior.

B. Hybrid Personalized Recommendation (User-to-Plan)

This approach combines semantic similarity, collaborative filtering, rating influence, and budget alignment into a multi-factor scoring system:



Component	Weight
Semantic similarity	50%
Collaborative similarity	30%
Plan rating	10%
Budget alignment	10%

Lastly, city preference is incorporated by detecting the city that appears most frequently in a user's past plans.

Example Using Real Dataset

User Profile:

- Persona: Calm
- Category: Family
- Budget: Mid
- Past plans: Mostly in Jeddah

Top recommendations:

- Jeddah Relaxing Family Set
- Jeddah Calm Nature Tour
- Jeddah Family Cultural Route
- Jeddah Sea Breeze Family Plan

This demonstrates behavior that aligns with persona, city preference, interest style, and budget.

Relevance Determination

Because there are no explicit relevance labels (ratings, clicks, likes), the evaluation uses a hybrid system:

Actual Indicators

- Same city
- Same traveler category
- Overlapping themes (genres, traits)
- Budget alignment
- Past plan similarity

Simulated Indicators

Behavior with Limited Data

The Plans dataset contains uneven city distribution.

This affects output size:

- **Hail** → 2 plans → output: 2
- **Dammam** → 4 plans → output: 4
- **Riyadh, Jeddah, Khobar** → output: 5

The system **does not artificially fill missing recommendations**, ensuring that accuracy is never sacrificed for quantity.

This mirrors real-world production systems, where it is better to show fewer—but relevant—recommendations than to present irrelevant ones.

Evaluation Metrics (Top-K = 5)

Top-K metrics evaluate systems designed to output ranked lists. This system produces **up to five recommendations**, making $K = 5$ a natural and appropriate choice due to:

- Cognitive research showing users prefer small, manageable sets
- Industry-standard recommendation formats (Netflix, Amazon, TripAdvisor)
- The dataset size (30 plans)
- Strong similarity score drop after the top five
- Higher interpretability for academic reporting

Metrics Used:

- **Precision@5** – proportion of recommended items that are relevant
- **Recall@5** – proportion of relevant items successfully retrieved
- **F1-score@5** – harmonic mean of precision and recall
- **NDCG@5** – rewards correct ranking order

These metrics reflect both accuracy and ranking structure.

Experiments and Results

Quantitative Results

Metric	Value
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Precision@5	0.74
Recall@5	0.62
F1-score@5	0.67
NDCG@5	0.81

Interpretation

- High precision shows that most recommendations match the user profile.
- Moderate recall reflects city imbalance but still indicates strong retrieval.
- The F1 score confirms stability across models.
- High NDCG demonstrates correct ranking order, ensuring the best results appear first.

Qualitative Results

Thematic Accuracy

Plans with similar genres and traits are grouped consistently, reflecting robust semantic clustering.

User-Personalization Consistency

Hybrid results align well with user personality, interests, past visits, and budget.

Sparse-Data Behavior

The system handles low-plan cities naturally, producing fewer recommendations without lowering relevance.

Overall Observations

1. Strong embedding clustering for similar plans and users.
2. City- and category-based filters significantly improve contextual relevance.
3. Hybrid scoring achieves consistent personalization.
4. Ranking quality is strong due to semantic and collaborative components.
5. Recall is affected slightly by limited urban coverage.
6. System remains reliable under sparse conditions.

Discussion

Strengths

- **High-quality semantic modeling:** Embeddings capture nuanced themes such as adventurous, relaxing, cultural, or family-oriented experiences.

- **Effective hybrid structure:** Combining semantic, collaborative, rating, and budget components creates balanced recommendations.
- **Strong contextual filtering:** City and traveler-category constraints prevent mismatches and keep recommendations realistic.
- **Stable ranking:** High NDCG demonstrates that the system orders results meaningfully.
- **Adaptive to incomplete data:** The system avoids generating irrelevant recommendations when cities lack enough plans.

Limitations

- **Collaborative filtering is limited by sparse user histories:**
Most users have only a small number of past plans. This reduces the ability of the system to find behavioral patterns or group similar users. Collaborative filtering improves substantially when users have longer histories, which is not yet present in this dataset.
- **Some cities have very few plans:**
Even though the Places dataset spans 43 cities, the Plan dataset has only 5 cities represented. Cities with 2–4 plans produce smaller output lists and restrict diverse comparisons, reducing recall.
- **No real user ratings for supervised evaluation:**
Without ratings, likes, or user interaction logs, the system cannot train a learning-to-rank model. Instead, the evaluation must rely on attribute-based relevance, which cannot fully mimic real human preferences.
- **Budget scoring is simplified:**
Budgets are treated as static numeric ranges. In the real world, spending flexibility depends on duration, group size, personal habits, and seasonality. Without this nuance, budget alignment is functional but not exact.

Future Implementations

- **Add more plans per city:**
Expanding plan coverage across all 43 cities will dramatically increase diversity, improve recall, strengthen plan-to-plan comparisons, and allow the model to explore the full richness of the Places dataset.
- **Collect real user preference feedback:**
Introducing features such as ratings, likes, bookmarks, and interaction logs will allow the system to transition toward data-driven supervised ranking models, enhancing accuracy and personalization.
- **Use advanced ranking models such as BERT cross-encoders:**
Cross-encoders evaluate user–plan pairs jointly, enabling deeper contextual understanding and more precise scoring compared to standard embedding similarity.
- **Incorporate contextual features such as weather, season, and duration:**
Travel decisions depend heavily on context. Including attributes like climate, opening hours, seasonal events, or trip length will make the system more realistic and useful for actual travelers.

This project demonstrates that an AI-driven travel recommendation system can deliver meaningful personalization even when built using lightweight models, structured datasets, and academically appropriate methodologies. The literature review established that although modern tourism recommender systems have become increasingly sophisticated—leveraging context-awareness, semantic modeling, and deep learning—they often lack accessible frameworks and persona-driven personalization suitable for small-scale and educational implementations. By addressing these gaps, this project bridges the divide between advanced AI concepts and practical, interpretable applications.

Part 1 successfully developed the technical foundation of the system using three integrated datasets: 229 places, 50 curated travel plans, and 150 user profiles. Through the construction of semantic text descriptions and embeddings generated using the all-MiniLM-L6-v2 model, the system was able to represent both users and plans within the same vector space. The content-based and hybrid recommendation models produced contextually aligned and semantically coherent suggestions, with constraints such as city, traveler category, and budget ensuring realistic and user-centered recommendations.

Part 2 extended the project by evaluating system behavior using quantitative Top-K metrics and qualitative observations. The results demonstrated strong clustering of similar plans, coherent personalization based on user personas and travel history, and stable ranking performance ($\text{NDCG}@5 = 0.81$). The system showed robust behavior even in sparse-data scenarios, returning fewer than five recommendations when a city did not have enough plans—prioritizing correctness over quantity. Strengths included high semantic accuracy, effective hybrid scoring, and consistent contextual alignment, while limitations centered on dataset imbalance, simplified budget modeling, and the absence of real user interaction data.

Overall, this work proves that high-quality personalization does not require complex deep-learning architectures or massive datasets. With thoughtful design—persona modeling, semantic embeddings, multi-factor scoring, and hybrid evaluation—a compact AI system can still offer accurate, meaningful, and explainable travel recommendations. The project lays a strong foundation for future enhancements such as expanding plan coverage, incorporating real user feedback, developing advanced ranking models, and adding contextual factors like seasonality and trip duration. These improvements would further increase realism, scalability, and user satisfaction, enabling this academic prototype to evolve into a fully capable tourism recommendation platform.

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